UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Formulas and symbols

Every formula in scope, then a glossary of every symbol used.

Class 011 (Legendre) · Legendre's and Cauchy's linear differential equations · compiled 17 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S011L·8	Unit 2 deck 011 “Legendre's differential equation” (SESSION 9), slide 8 (rendered .pptx)
T011L 00:05:20	AV-summary transcript “Legendre differential equation” (21.9 min, listed first), segment start
T011Lb 00:12:26	AV-summary transcript “Legendre differential equations_Continued” (23 min, second), segment start
CW U2 P30	Unit 1 & 2 class-work sheet, Unit 2 problem 30
ESA Jul-23 3a	Past paper: UE22MA141A July 2023, question 3.a (Jan-24 s1...s4 = UE23MA141A Jan 2024 sets 1–4)

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A Formula sheet

1 Linear equations with variable coefficients: Legendre's and Cauchy's forms

E1.1 · Legendre's equation of order n

ESA ★★★

$$k_0(ax+b)^n\frac{d^ny}{dx^n}+k_1(ax+b)^{n-1}\frac{d^{n-1}y}{dx^{n-1}}$$
$$+k_2(ax+b)^{n-2}\frac{d^{n-2}y}{dx^{n-2}}+\cdots+k_ny=X$$

GIVES The form to recognise.

APPLIES Power of $(ax+b)$ = order of the derivative in every term; $k_0\neq 0$.

NOTE Slide 011L·5 prints the third derivative as $\frac{d^{n-1}y}{dx^{n-1}}$: it is $\frac{d^{n-2}y}{dx^{n-2}}$.

S011L·4 · S011L·5 · T011L 00:01:38 · T011L 00:04:06

E1.2 · Cauchy's equation of order n $(a=1,\ b=0)$

ESA ★★★

$$k_0x^n\frac{d^ny}{dx^n}+k_1x^{n-1}\frac{d^{n-1}y}{dx^{n-1}}+k_2x^{n-2}\frac{d^{n-2}y}{dx^{n-2}}+\cdots+k_ny=X$$

GIVES The special case.

APPLIES Coefficient x^k in front of the k-th derivative.

NOTE Slide 011L·4 also prints $\frac{d^{n-1}y}{dx^{n-1}}$ in the third term; it is $\frac{d^{n-2}y}{dx^{n-2}}$.

S011L·4 · T011L 00:16:51 · T011L 00:17:52 · ESA Jan-24 s3 3d

2 The substitution $ax+b=e^z$ and the operator results

E2.1 · The substitution and $\frac{dz}{dx}$

ESA ★★★

$$ax+b=e^z,\qquad z=\log(ax+b),\qquad x=\frac{e^z-b}{a},\qquad \frac{dz}{dx}=\frac{a}{ax+b}$$

GIVES The change of variable.

APPLIES Every Legendre equation.

NOTE Cauchy: $x=e^z$, $z=\log x$, $\frac{dz}{dx}=\frac{1}{x}$.

S011L·5 · T011L 00:05:20 · T011L 00:06:11 · T011Lb 00:01:49

E2.2 · First derivative

ESA ★★★

$$\frac{dy}{dx} = \frac{dy}{dz}\cdot\frac{a}{ax+b} \quad\Rightarrow\quad (ax+b)\frac{dy}{dx} = aDy, \qquad D = \frac{d}{dz}$$

GIVES The first-order term.

APPLIES Chain rule.

NOTE Cauchy: $x\frac{dy}{dx} = Dy$.

S011L·6 · T011L 00:07:16 · T011L 00:07:44 · T011L 00:18:54

E2.3 · Second derivative

ESA ★★★

$$(ax+b)^2\frac{d^2y}{dx^2} = a^2D(D-1)y = a^2(D^2-D)y$$

GIVES The second-order term.

APPLIES Product rule on $\frac{a}{ax+b}\frac{dy}{dz}$, then the chain rule.

NOTE Cauchy: $x^2\frac{d^2y}{dx^2} = D(D-1)y$. Not a^2D^2y.

S011L·6 · T011L 00:08:53 · T011L 00:11:45 · T011L 00:19:07

E2.4 · Third derivative and the general rule

ESA ★★

$$(ax+b)^3\frac{d^3y}{dx^3} = a^3D(D-1)(D-2)y$$
$$(ax+b)^n\frac{d^ny}{dx^n} = a^nD(D-1)(D-2)\cdots(D-n+1)y$$

GIVES Higher-order terms (n linear factors).

APPLIES Order 3 and 4 equations (lecture examples).

NOTE Slide 011L·6 shows $n = 1, 2, 3$; the general line is added for clarity. Cauchy: $x^4y'''' = D(D-1)(D-2)(D-3)y$.

S011L·6 · T011L 00:12:21 · T011L 00:12:34 · T011Lb 00:20:04

E2.5 · Second-order Legendre equation reduced

ESA ★★★

$$(ax+b)^2y'' + k_1(ax+b)y' + k_2y = X$$
$$\Longrightarrow\ \left(a^2D(D-1) + k_1aD + k_2\right)y = X\left(\tfrac{e^z-b}{a}\right)$$

GIVES A constant-coefficient equation in z.

APPLIES Every ESA question.

NOTE Divide by a^2 if convenient; the AE is $a^2m(m-1) + k_1am + k_2 = 0$.

S011L·7 · T011L 00:14:20 · T011L 00:15:57 · ESA Jul-23 3a

E2.6 · Back to x ESA ★★★

$$e^{mz} = (ax+b)^m, \qquad z = \log(ax+b), \qquad \cos\beta z = \cos(\beta\log(ax+b))$$

GIVES The final answer in x.

APPLIES Last step, always.

NOTE Repeated root m: $(c_1 + c_2 z)e^{mz} = (ax+b)^m[c_1 + c_2\log(ax+b)]$.

S011L·9 · T011Lb 00:09:11 · T011Lb 00:12:12 · T011Lb 00:17:39

3 Legendre problems: the slide example, CW P29–P30 and five ESA questions

E3.1 · What the right-hand side becomes ESA ★★★

$$x = \tfrac{e^z - b}{a}: \quad (ax+b)^k \to e^{kz}, \quad px+q \to \tfrac{p}{a}(e^z - b) + q$$
$$\log(ax+b) \to z, \qquad \sin(k\log(ax+b)) \to \sin kz$$

GIVES The PI type in z.

APPLIES Before the PI.

NOTE A polynomial in x gives a sum of e^{kz} and a constant (Type 1 with $k = 0$).

S011L·8 · T011Lb 00:14:09 · CW U2 P30 · ESA Jun-24 3b

E3.2 · PI rules needed in z (classes 003–004) ESA ★★★

$$\frac{1}{f(D)}e^{kz} = \frac{e^{kz}}{f(k)},\ f(k)\neq 0; \qquad \frac{1}{f(D)}e^{kz} = \frac{z\,e^{kz}}{f'(k)},\ f(k)=0\neq f'(k)$$
$$\frac{1}{(D-k)^2}e^{kz} = \frac{z^2}{2}e^{kz}, \qquad \frac{1}{D^2+\beta^2}\sin\beta z = -\frac{z}{2\beta}\cos\beta z$$

GIVES Every PI in Topic 3.

APPLIES Failure cases in 4 of 6 ESA questions.

NOTE A constant c is ce^{0z}: $\frac{c}{f(0)}$.

ADDED FOR CLARITY T011Lb 00:10:29 · T011Lb 00:16:27 · ESA Jul-23 3a · ESA Jan-24 s2 3c · CW U2 P29

E3.3 · ESA answers (Legendre) ESA ★★★

$$\text{Jul-23: } y = (2x+1)^2\big[c_1 + c_2\log(2x+1) + \log^2(2x+1)\big]$$
$$\text{Jan-24 s2: } y = (x+2)\big[c_1 + c_2\log(x+2)\big] + \tfrac{3}{2}(x+2)\log^2(x+2) - 2$$

GIVES The two repeated-root ESA answers.

APPLIES W3.5, W3.7.

NOTE Others: W3.3, W3.6, W3.8.

ESA Jul-23 3a · ESA May-24 3a · ESA Jan-24 s2 3c

4 Cauchy problems: lecture examples, CW P31–P33 and ESA Jan-24 s3 3d

E4.1 · Cauchy: right-hand side in z ESA ★★★

$$x = e^z: \quad x^m \to e^{mz}, \qquad \frac{1}{x} \to e^{-z}, \qquad \log x \to z, \qquad x^m \log x \to z\,e^{mz}$$

GIVES The PI type.

APPLIES Every Cauchy problem.

NOTE $x^4 \sin x \to e^{4z}\sin(e^z)$ is not a standard form (E4.2).

T011Lb 00:06:06 · T011Lb 00:06:23 · CW U2 P31 · ESA Jan-24 s3 3d

E4.2 · Non-standard right-hand side: one factor at a time (class 009) ESA ★★

$$\frac{1}{D-m}X = e^{mz}\int X\,e^{-mz}\,dz$$

GIVES The PI of $(D-m_1)(D-m_2)y = X$ in two integrations.

APPLIES ESA Jan-24 s3 3d ($e^{4z}\sin e^z$).

NOTE Or divide by x^2 and use variation of parameters in x (class 011 VoP).

ADDED FOR CLARITY ESA Jan-24 s3 3d · ESA Jan-24 s4 3c

B Symbol glossary

Symbol	Name	Meaning	Appears in
a, b	Constants of the linear factor	$ax+b$ in Legendre's equation; Cauchy: $a=1$, $b=0$	E1.1, E2.1
$k_0, \dots, k_n$	Coefficients	Constants multiplying each term; $k_0 \neq 0$	E1.1, E1.2, E2.5
X	Right-hand side	Function of x (capital); rewritten in z after the substitution (slide: $f(z)$)	E1.1, E2.5
z	New variable	$z=\log(ax+b)$ or $z=\log x$; independent variable after the substitution	E2.1
D	Operator	$\frac{d}{dz}$ after the substitution (clash: $\frac{d}{dx}$ in classes 002–009); some books θ	E2.2–E2.4
n	Order	Highest derivative; also the power of $(ax+b)$ on it	E1.1, E2.4
m	AE root	Root of the auxiliary equation in z; $e^{mz}=(ax+b)^m$	E2.6
k	Exponent of the RHS	e^{kz} terms of the right-hand side (clash with the coefficients k_i)	E3.1, E3.2
y_c, y_p	CF and PI	Complementary function and particular integral; $y=y_c+y_p$	W3.1
$\log$	Natural logarithm	$\log_e$; needs $ax+b>0$ ($x>0$ for Cauchy)	E2.1, E2.6

Remark. z is the new independent variable: $z=\log(ax+b)$ (Legendre) or $z=\log x$ (Cauchy). From the substitution on, D means $\frac{d}{dz}$, not $\frac{d}{dx}$ (the lecture stresses this; some books write θ). Capital X is the right-hand side; small x the variable. The deck writes the coefficients $k_0, k_1, \dots, k_n$; a, b are the constants of the linear factor $ax+b$. $\log$ means $\log_e$, and the solutions hold where $ax+b>0$ (Cauchy: $x>0$).